

Playwright Test Automation

Overview

Playwright is a modern end-to-end testing framework developed by Microsoft. It supports Chromium, Firefox, and WebKit browsers and enables reliable, fast automation for web applications with a single API.

Key Features

- Cross-browser support: Chromium, Firefox, WebKit.
- Auto-wait mechanism — no need for manual sleeps.
- Supports Python, JavaScript, Java, and .NET.
- Built-in tracing, screenshots, and video recording.
- Handles iframes, tabs, popups, and file uploads natively.

Basic Example (Python)

```
from playwright.sync_api import sync_playwright

with sync_playwright() as p:
    browser = p.chromium.launch(headless=True)
    page = browser.new_page()
    page.goto('https://example.com')
    print(page.title())
    browser.close()
```

Locators and Actions

Playwright uses smart locators to find elements reliably. It supports CSS selectors, XPath, text selectors, and role-based locators which align well with accessibility standards.

Common Interactions

```
# Click a button
page.get_by_role('button', name='Submit').click()

# Fill a form field
page.get_by_label('Email').fill('test@example.com')

# Assert text on page
expect(page.get_by_text('Success')).to_be_visible()

# Take a screenshot
page.screenshot(path='screenshot.png')
```

Why Playwright over Selenium?

- Faster execution due to protocol-level browser communication.
- Auto-waits eliminate flaky tests caused by timing issues.
- Built-in network interception and mocking.
- Parallel test execution out of the box.